YOLD: You Only Look Denseness for Tiny Object Detection in Aerial Images

Guangshuai Gao, *Member, IEEE*, Yunqi Shang, Junyu Gao, *Member, IEEE*, Haojie Guo, Yan Dong, and Chunlei Li

***Abstract*—Aerial image object detection has gained increasing attention recently. Compared to natural scenes, this task is more challenging due to two main factors: 1) a large number of densely packed tiny objects with blurred appearances that hinder detection accuracy and 2) nonuniform data distributions that limit detection efficiency. To address these challenges, we present you only look denseness (YOLD), a simple yet efficient detection framework. Specifically, for dense small objects, we propose an end-to-end anchor-free detector guided solely by point supervision, improved from the classical CenterNet, the key difference of which lies in using localization-based distance transform maps instead of heatmaps. To tackle nonuniform data distribution, a straightforward and effective dense region extraction (DRE) module is designed to identify clustered regions containing challenging instances such as dense tiny objects, thereby improving detection efficiency. Additionally, we design the regression loss by incorporating the generalized Euclidean distance (GED) with an adaptive weighted mechanism based on objects' areas, which improves the detection accuracy of tiny object detection. Finally, extensive experiments are carried out on three publicly available large-scale aerial image object detection datasets, namely, VisDrone, UAVDT, and DTOD. The results confirm both the effectiveness of the proposed approach and its superiority over state-of-the-art methods.**

***Index Terms*—Aerial image object detection, dense region extraction (DRE), distance transform maps, nonuniform data distribution, tiny objects.**

I. Introduction

OBJECT detection is a form of multitask learning that simultaneously estimates object locations and categories. As a fundamental problem in computer vision, it has a wide range of applications, including disaster prediction and traffic management. Representative detection paradigms (e.g., Faster R-CNN [1], YOLO [2], and SSD [3]) perform well on natural image datasets (e.g., MS COCO [4] and Pascal VOC [5]). However, their performance on aerial images is limited by two main challenges.

1) Severe scale variation, particularly the prevalence of extremely small objects, which occupy only a few pixels in high-resolution images (e.g., 2000×1500 in VisDrone [6]), limits the upper bound of performance improvement. This makes small object detection particularly challenging under severe scale variation, as tiny objects are easily overwhelmed by background patterns in deep features, leading to weak signals and frequent missed detections.
2) Sparse clustered and nonuniform data distribution, such as vehicles are often concentrated at intersections and pedestrians gather in public squares, whereas most other regions exhibit sparse object distributions or are dominated by background, thereby complicating detection and causing inefficient background processing.

Compared with objects in natural images, the scale challenge causes less effective feature representation of deep networks for objects in aerial images [7], [8]. Therefore, it is difficult for modern detectors to effectively leverage appearance information to distinguish the objects from the surrounding background or similar objects. To alleviate the above issues, a simple and straightforward approach is the uniform cropping [9], [10], as illustrated in Fig. 1(a), which divides the image into several evenly sized chips and performs detection on each individual chip. It is easy to implement and can partially alleviate the challenge posed by limited resolution for tiny object detection. However, the sparsity of object distribution is overlooked. Consequently, a significant amount of computational resources is wasted on regions with few or no objects, leading to slower inference speed and reduced detection accuracy.

Recently, mainstream solutions are based on clustered regions search strategies [11], [12], [13], [14], [15], which methods usually design dedicated schemes to search the clustered regions that include large number of tiny objects, thereby increasing the utilization rate of tiny objects and enhancing the detection speed. ClusDet [11] proposes a clustered detection framework. This framework incorporates a cluster proposal network (CPNet) and a scale estimation network (ScaleNet). Specifically, CPNet reduces the number of image partitions to

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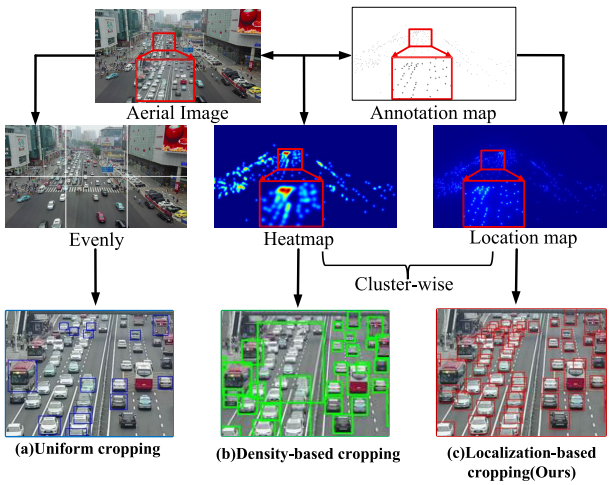


Fig. 1. Comparison between uniform cropping, density cropping, and our proposed location map cropping strategy. (a) Uniform cropping. Many cropped patches contain redundant background regions with few or no objects, leading to inefficient computation. (b) Density-based cropping. Blurred Gaussian blobs provide coarse outlines of dense regions, which may lead to missing detection, especially for dense tiny objects. (c) Our proposed localization-based cropping strategy. Clearer location maps provide finer object boundaries, which effectively identify and extract clustered regions to achieve a better tradeoff between accuracy and efficiency.

improve detection efficiency. ScaleNet leverages cluster-based scale estimation with context modeling, which significantly enhances the performance of tiny object detection. However, the reliance on additional subnetworks increases model complexity and computational cost. DMNet [12] draws inspiration from crowd counting to generate density maps for reflecting object distribution and crops clustered regions via density thresholds. Albeit adaptive, its performance depends heavily on the accuracy of the map, which may degrade in sparse or occluded scenarios. YOLC [16] generates density maps using Gaussian responses and enhances the regression branch with deformable convolution to better capture tiny object details. It reduces computation and improves multiclass tiny object detection. However, blurred boundaries or severe overlaps can cause ambiguous heatmap responses, affecting dense region localization and detection accuracy, as illustrated in Fig. 1(b). These strategies show promising results, since the clustered regions are preserved while the backgrounds are suppressed as much as possible. However, they obtain the clustered regions in a coarse manner, which may hinder the lift of final detection performance.

Motivated by the above observation, we propose a sibling work of YOLC [16], YOLD, an anchor-free localization-based clustered network for aerial image object detection. YOLD is developed based on CenterNet [17], with heatmaps replaced by location maps, as illustrated in Fig. 1(c). It consists of three key components: a location map generation (LMG) module, a dense region extraction (DRE) module, and an optimized training loss strategy. Specifically, LMG employs a robust Gaussian-weighted distance transform map (GDTM) to adaptively generate location maps according to object density, enabling more precise localization of densely packed tiny objects. Additionally, for the nonuniform data distribution, DRE adaptively searches clustered object regions and rescales them to suitable scales for the detectors. Compared to previous strategies [11], [12], [15], it is an unsupervised module that does not require a dedicated CPNet, simplifying implementation and improving inference efficiency. Furthermore, to enhance detection accuracy and guarantee stable training, we introduce a generalized Euclidean distance (GED) loss based on Mahalanobis distance (MD) to accommodate scale and directional variations, combined with an L_1 loss to prevent gradient vanishing in large objects, thereby ensuring robust training and precise localization.

In summary, the contributions of this work are threefold.

1) An effective and efficient localization-based anchor-free object detector named YOLD is introduced, featuring a streamlined and elegant architecture, in which a GDTM is employed to adaptively generate location maps according to object density, enabling more accurate localization of densely packed tiny objects.
2) An unsupervised DRE module is introduced to adaptively search the clustered regions, which is economic and lightweight.
3) A novel GED-based regression loss is proposed, which extends MD to enhance adaptability to scale variations and directional characteristics while remaining stable, differentiable, and particularly effective for tiny objects.

The remainder of this article is organized as follows. Section II reviews some literature closely related to our work. The proposed approach is elaborately introduced in Section III. Experimental results and analysis are provided in Section IV. Finally, this article is concluded in Section V.

II. Related Work

A. General Object Detection

General object detection can be roughly classified into two categories: anchor-based and anchor-free methods. According to whether generating candidate proposals, anchor-based models can be divided into two-stage and single-stage methods. Representative two-stage methods are R-CNN series [1], [18], [19], Cascade RCNN [20], and R-FCN [21]. These approaches, which involve generating region proposals followed by classification and bounding box regression, offer high accuracy but relatively low efficiency. To speed up the training, single-stage detectors are proposed to regress the bounding boxes and classify objects directly. Classical methods include YOLO series [22], SSD [3], and RetinaNet [23]. Bypassing complicated handcrafted anchors, some anchor-free models are presented. They are divided into dense box- and keypoint-based methods. Typical dense box-based methods are Foverbox [24] and FCOS [25]. They usually predict the distances between locations to the four sides of the bounding boxes. Keypoint-based methods only predict the corner or center points of the objects, which mainly include CornerNet [26], Cornernet-lite [27], and CenterNet [17], [28]. These excellent object detectors show promising performance in natural scenes; however, they may obtain unsatisfactory results when applied to aerial images.

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B. Aerial Image Object Detection

Aerial image object detection faces more challenges than general object detection, due to various types of acquisition platforms, different shooting angles, a wide field of view, and so on. More importantly, a large number of tiny objects and shifting data distribution limit the upper bound of detection performance. To address these issues, image cropping strategies have been proven to be the most effective tools. Unel et al. [13] provide a tiling-based approach that demonstrates the power of uniform cropping for tiny object detection. Despite being effective, the data distribution in aerial images makes the uniform image crops contain much background information. In addition, it may cut off the large objects into two or more parts. Based on this idea, ClusDet [11] improves the detection efficiency by focusing the detector on the clustered regions. It presents a CPNet to obtain a set of clustered regions and then provides a ScaleNet to rescale the clustered regions to fit the detectors. Finally, it fuses the detection results from the rescaled clustered regions and the original image. Furthermore, according to the distribution of objects, DMNet [12] and CDMNet [29] exploit density maps to represent object presence and infer appropriate scale information from density intensities. GLSAN [15] introduces a global–local fusion strategy within a progressive multiscale network to achieve precise detection. Although these methods identify dense object regions, the extracted regions often contain coarse object boundaries, which limit suboptimal detection results.

YOLC [16] presents a local scale module (LSM) to search the clustered regions. It computes object density according to the object number within each grid and then ranks the grids in descending order to adaptively localize the top-K dense regions, which are selected as candidate crops to enable focused detection on densely clustered regions. YOLC [16] still suffers from two main limitations: 1) the generated density maps often produce overlapping blurry blobs, which may obscure object centers and lead to missed detections; and 2) grid partitioning may split a truly dense region into multiple subregions, leading to the loss of real clustered areas. In contrast, YOLD addresses these limitations by leveraging localization-based guidance to provide more precise object positions. We further design a lightweight unsupervised DRE module to adaptively search clustered regions and introduce an optimized GED loss to ensure stable model training.

III. Methodology

A. Preliminary

CenterNet [17] considers each object as a keypoint in the image and uses the point as the center point to predict and regress the sizes, directions, poses of the objects, and so on, which is a simple and efficient anchor-free detection framework. Therefore, object detection can be formulated as a keypoint regression problem. Specifically, the model generates a heatmap (also known as density map) from a fully convolutional network, and object centers are obtained by identifying local maxima in the heatmap, from which object sizes can be inferred.

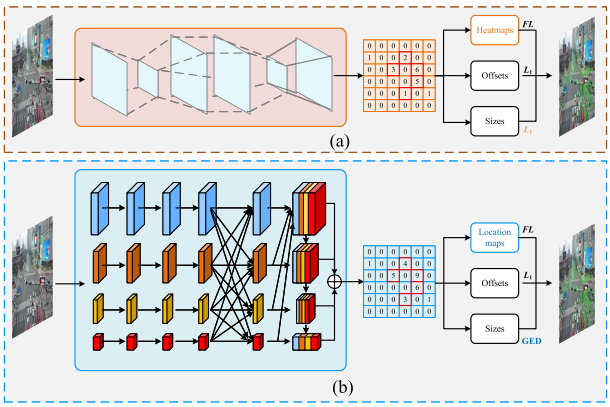


Fig. 2. Comparison of network architectures between (a) CenterNet [17] and (b) our proposed YOLD (bottom row). Three components are improved, including backbone network, regression manner, and the loss function for size prediction, which can be viewed in (a) orange and (b) blue boxes and fonts.

Inspired by dense object localization methods (e.g., crowd localization [30], [31], [32]), the variation in density intensity can indicate the probability of object presence, which is useful for image cropping or extracting regions of interest in aerial imagery. CenterNet [17] directly estimates object density maps, consistent with this observation. However, directly applying CenterNet [17] to aerial images may lead to suboptimal performance due to the large number of closely clustered tiny objects. Specifically, density maps often fail to provide precise localization for each object, primarily due to two factors: 1) density maps consist of a series of blurred Gaussian blobs and 2) severe overlaps occur in densely populated regions.

To address the above issues, we replace the heatmaps with location maps of all object center positions and propose a framework termed YOLD. This design enables the model to concatenate clustered regions and follows a pipeline similar to CenterNet [17] while differing in backbone design, regression strategy, and loss functions, as illustrated in Fig. 2. Specifically, in CenterNet [17], Hourglass networks are employed as the backbone. However, considering the large number of tiny objects in aerial images and the potential information loss caused by multiple convolution and pooling layers, we employ a high-resolution feature extraction backbone, HRFPN [33]. The aerial image is first fed into the backbone, and a location map is generated to replace the conventional heatmap, enabling precise prediction of object center positions. Then, other properties such as sizes, orientations, and poses can be easily inferred from the center locations. In addition, the conventional L_1 loss used in vanilla CenterNet [17] is suboptimal for tiny object regression. To better capture fine-grained localization and size information, we propose a novel GED loss function for stable convergence training optimization.

B. You Only Look Denseness

As illustrated in Fig. 3, the overall architecture of our proposed YOLD framework includes three main components: 1) LMG; 2) DRE; and 3) GED loss function.

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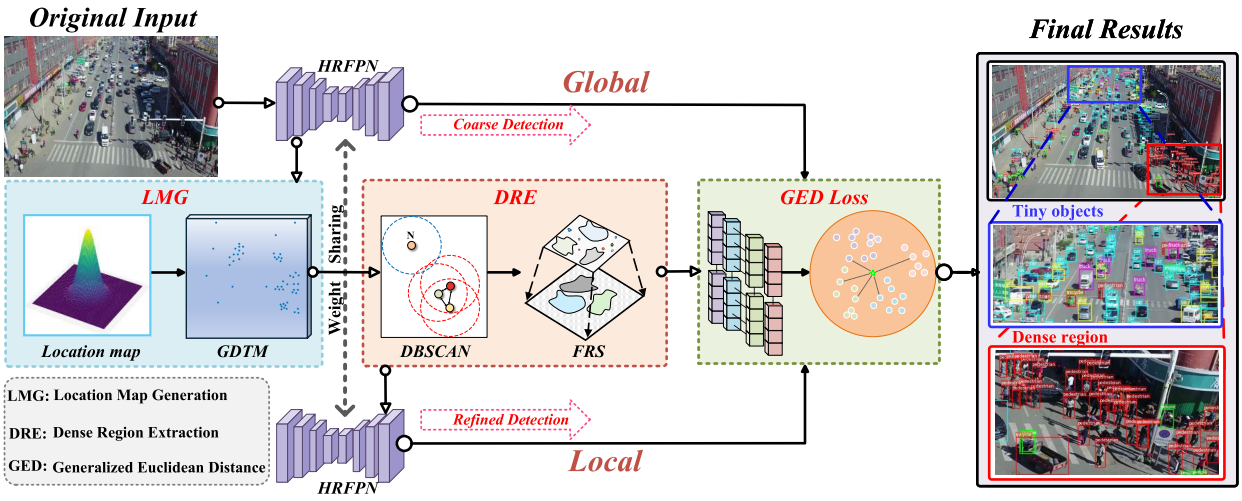


Fig. 3. YOLD framework with the clustered region mining module. The final detection results are obtained from the combination of the image crops and global images.

YOLD adopts a coarse-to-fine detection paradigm, in which the process begins with LMG to generate a location map to encode the center positions of objects within the input image, serving as an initial estimation of object distribution. Based on the location map, the DRE module adaptively identifies and crops high-density regions containing clusters of objects. Subsequent fine-grained detection is conducted on these localized regions using a shared-weight detector. The final detection results are obtained by fusing the coarse predictions from the global image and the refined predictions from the local dense regions. To further improve localization accuracy, especially in densely populated scenes with tiny objects, we introduce the GED loss and combine it with a Smooth L_1 regression term to replace the conventional L_1 loss. This loss design more effectively models the spatial and geometric relationships between objects while providing more stable gradients during optimization, leading to improved localization precision and overall detection performance. Additionally, we adopt the decoupled detection head design from YOLC [16] and integrate ASPP [34] and CBAM [35] modules. These enhancements improve object localization accuracy and also enhance multiscale feature fusion and contextual information modeling, thereby effectively boosting the detection performance for small objects.

C. Location Map Generation

We first define the Euclidean distance transform map as follows:

$$D(x,y) = \min_{(x',y')\in\mathcal{S}} \sqrt{(x-x')^2 + (y-y')^2} \tag{1}$$

where $\mathcal{S}$ represents the set of ground truth (GT) center points and $D(x,y)$ denotes the distance from a pixel (x,y) to the nearest object's center.

The Euclidean distance is simple, computationally efficient, and widely used as a standard metric to measure similarity between points. However, the Euclidean distance suffers from a scale invariance problem, which may lead to similar distance values for both nearby and remote objects, thereby affecting localization accuracy. Moreover, Euclidean distance performs poorly in high-dimensional spaces, especially when the range of distances varies greatly, making direct regression challenging.

To address these issues, we propose a more robust GDTM, defined as follows:

$$\mathrm{GD}(x,y) = e^{-\gamma(D(x,y))^2} \tag{2}$$

where γ is an adaptive decay factor controlling the spread of the response.

Most localization methods employ a fixed decay factor, which struggles with significant density variations [36]. To address this, we introduce an adaptive density-aware decay strategy. Specifically, we utilize a KD-Tree [37] to efficiently model spatial distributions and assess local object density through fast nearest neighbor queries. By computing the Euclidean distances between each object and its neighbors using the KD-Tree, we dynamically adjust the decay factor γ to better reflect local object density. This allows denser regions to use a larger decay factor (resulting in sharper peaks), while sparser regions use smaller ones (yielding broader responses), thus better adapting the location map to density variations.

Concretely, we first compute the average distance between each object and its $k-1$ nearest neighbors

$$D_i = \frac{1}{k-1} \sum_{j=1, j\neq i}^{k} d\left(C_i, C_j\right) \tag{3}$$

where $d(C_i, C_j)$ is the Euclidean distance between object i and object j; C_i and C_j denote the center coordinates of objects i and j, respectively; and k is the number of neighbors, with a default value of 4.

To ensure consistency across different scenes, we normalize the average distance

$$\hat{D}_i = \frac{D_i}{D_{\max}} \tag{4}$$

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where $\hat{D}_i$ is the normalized average distance for object i and $D_{\max}$ is the maximum distance among all object pairs in the image.

The decay factor γ_i for any object is then computed as follows:

$$\gamma_i = \min\left(\frac{\alpha}{\hat{D}_i + \delta}, \gamma_{\max}\right) \tag{5}$$

where α is a scaling factor (empirically set to 10) that controls the global decay rate, $\gamma_{\max} = 5$ is the maximum allowable decay to prevent overly weak responses, and $\delta = 10^{-6}$ is a small constant introduced to ensure numerical stability by preventing division by zero.

This density-aware strategy allows sharper and more localized responses in dense regions, and broader, more tolerant responses in sparse regions. As a result, the generated location maps are more adaptive to varying object densities, significantly improving detection performance in complex scenes with nonuniform distribution.

D. Dense Region Extraction

Except for a large number of highly congested small objects in aerial images, there exists a severe data imbalance. In other words, some object instances are gathering into clusters, while others are distributed in sparse scatters. To address these challenges and detect objects in high-resolution images with limited computational resources, the most mainstream method is to perform local refinement detection.

However, direct random cropping or uniform cropping results in a large amount of invalid background redundancy, mainly because these methods are unable to accurately and efficiently localize the key ROIs. DMNet [12] and CDMNet [29] use the number of objects to define dense clusters, which usually rely on fixed statistical windows (e.g., grid divisions and sliding windows). Meanwhile, they are unable to reflect the localized densities, i.e., the same number of objects in regions with different sizes may present completely different density distributions.

To mitigate this problem, we introduce a DRE module to adaptively seek the clustered regions and then enlarge them to the sizes adapting the object detectors. DRE consists of two main steps: subregion selection and foreground region stitching (FRS), as shown in Fig. 4. Specifically, a classical clustering algorithm, DBSCAN [38], is employed to identify dense subregions. Once the dense regions are located, the corresponding subregion images are mosaicked to form a new sample. To ensure compatibility with the detector, the scale and aspect ratio of the newly generated image are adaptively refined. The specific process of DRE is documented in Algorithm 1.

1) Subregion Selection: Given the location map generated from the coarse detection process, an empirical threshold is set to binarize the location maps of all categories to generate a location mask. In order to find clusters efficiently, we employ a clustering algorithm for unsupervised dense region search.

Different from the K-means strategy adopted in GLSAN [15], which requires a predefined number of clusters, our subregion selection is based on the DBSCAN clustering algorithm [38], which discovers clusters by identifying high-density regions and grouping density-connected samples, characterizing the closeness of sample distributions through a set of neighborhood factors, without requiring the number of clusters as input. It characterizes the density relationship among data points using two key parameters: the minimum number of points within a cluster (*minPts*) and the neighborhood radius (ϵ). In our implementation, *minPts* is set to 3, and the neighborhood radius ϵ is adaptively estimated using the KD-Tree employed to compute the decay factor γ in our GDTM. Specifically, for each image, we compute the distances from each object center to its k-nearest neighbors and set the median of these distances across all objects as ϵ. By relying on a unified notion of local density, both the response map generation and the clustering process are consistently aligned with the underlying spatial distribution of objects. This enables DBSCAN to automatically adjust to varying object densities across different scenes without manual parameter tuning. This strategy not only allows for the identification of clusters with arbitrary shapes but also effectively separates outliers, thereby improving detection efficiency.

After clustering, in order to achieve precise localization of the regions for local detection, we further perform boundary fitting on each valid cluster. Specifically, for the set of object points within each cluster, we apply the convex hull algorithm to enclose their boundaries and construct a compact polygonal region. This polygonal region effectively envelops all core points in the cluster while preserving the compactness and spatial consistency of the region to the greatest extent. Such a strategy provides a clear spatial constraint for subsequent local feature extraction and object regression. This boundary fitting not only enhances the geometric representation of dense regions but also reduces interference from redundant

Algorithm 1 Specific Process of DRE

Input: Location map M, foreground coordinates Q, image height H, image width W, number of subregions k
Output: Stitched dense region image I

$M' \leftarrow Binarize(M)$
$Q \leftarrow ExtractForeground(M')$
$\{P_1, P_2, \ldots, P_K\} \leftarrow DBSCAN(Q)$
$R \leftarrow \emptyset$
for each P_k in $\{P_1, \ldots, P_K\}$ **do**
  $R_k \leftarrow ConvexHull(P_k)$
  $R \leftarrow R \cup \{R_k\}$
end for
$R \leftarrow SortByArea(R, \text{descending})$
$\hat{R} \leftarrow TopK(R, k)$
for each R_i in $\hat{R}$ **do**
  $R_i \leftarrow Expand(R_i, \text{scale} = 1.2)$
  $I_i \leftarrow Crop(I_0, R_i)$
  $I_i \leftarrow Resize(I_i, \text{scale} = 1.5)$
end for
$I \leftarrow Concat(\{I_1, \ldots, I_k\}, \text{mode} = \text{compact})$
$I \leftarrow PadToSize(I)$
return I

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background information, thereby improving the accuracy and robustness of refined local detection.

The subregions predicted by the DRE module are derived from the output of the global coarse detector, which is a location map encoded based on the objects' center points. To prevent boundary truncation resulting from clustering that relies solely on center points, following [16], we enlarge the regions to 1.2 times, to moderately expand the cropped regions. This expansion ensures that the entire object can be completely enclosed in each local region.

2) Foreground Region Stitching: To balance detection accuracy and inference efficiency, we follow the empirical setting in YOLC [16] and limit the number of cropped subregions to a maximum of $N = 3$. However, each dense region (i.e., cropped patch) is processed independently by the fine detector, which leads to significant computational redundancy. To alleviate this, we adopt a mosaic-based stitching strategy inspired by PHSPPOG [39], merging multiple subregions into a single image to enable parallel inference. However, directly stitching too many patches may introduce redundant background and easily recognizable objects, which can reduce model sparsity and overall performance. Therefore, we sort subregions according to their areas and prioritize stitching the larger ones, ensuring a compact and informative composite image for efficient local refinement.

E. GED Loss

For the classification branch, we still retain the improved focal loss used in CenterNet [17], which effectively mitigates the issue of class imbalance between positive and negative samples.

For the regression branch, traditional bounding box regression losses, such as the L_1 loss or IoU-based losses, suffer from two fatal flaws: 1) directional characteristic of objects is not taken into account; and 2) lack of scale sensitivity, making it difficult to distinguish between objects of different sizes.

To address the above limitations, we introduce the MD to redefine the regression loss. This distance metric accounts for correlations between variables and adapts the Euclidean space through the inverse covariance structure. Given a positive-definite covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{d\times d}$, the MD between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ is defined to adapt the geometric space to match the underlying data distribution

$$d_{\mathrm{MD}}(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x}-\mathbf{y})^{\top} \mathbf{\Sigma}^{-1} (\mathbf{x}-\mathbf{y})}. \tag{6}$$

To enforce the model to learn distance functions tailored to specific characteristics such as object size or confidence, we further generalize this formulation through the GED, which replaces the inverse matrix in the MD with a learnable, symmetric positive-definite matrix $\mathbf{\Lambda} \in \mathbb{S}^d_{++}$

$$d_{\mathrm{GED}}(\mathbf{x}, \mathbf{y}; \mathbf{\Lambda}) = \sqrt{(\mathbf{x}-\mathbf{y})^{\top} \mathbf{\Lambda} (\mathbf{x}-\mathbf{y})} \tag{7}$$

where $\mathbb{S}^d_{++}$ denotes the set of all symmetric positive-definite matrices of size $d \times d$.

To handle large-scale variations in aerial object detection, we propose an *adaptive parameterization* of $\mathbf{\Lambda}$ that emphasizes tiny objects by scaling based on object area. The matrix $\mathbf{\Lambda}$ is defined as follows:

$$\mathbf{\Lambda} = \left(\mu \cdot \exp\left(-\frac{2\sqrt{A}}{\beta S}\right) + \nu\right) \mathbf{I} \tag{8}$$

where A denotes the object area, S is a normalization constant corresponding to the input image size (e.g., $S = 1024$), β controls the sensitivity to tiny objects, and μ and ν are stability parameters (we set $\mu = \nu = 1.0$ in our implementation).

To enable stable training, we define a differentiable loss function based on GED

$$L_{\mathrm{GED}} = \frac{\log(1 + d_{\mathrm{GED}})}{\tau + \log(1 + d_{\mathrm{GED}})}, \quad \tau \geq 1. \tag{9}$$

This loss function satisfies the following properties.

1) *Boundedness*: $L \to 0$ as $d_{\mathrm{GED}} \to 0$, and $L \to 1$ as $d_{\mathrm{GED}} \to \infty$.
2) *Smoothness*: Differentiable and numerically stable.
3) *Robustness*: The loss exhibits a sublinear (logarithmic) growth, providing robustness against outliers and preventing gradient explosion for large residuals.

To ensure numerical stability during computation, we implement the distance using *Cholesky decomposition*

$$d_{\mathrm{GED}}(\mathbf{x}, \mathbf{y}) = \|\mathbf{L}(\mathbf{x}-\mathbf{y})\|_2, \quad \mathbf{\Lambda} = \mathbf{L}\mathbf{L}^{\top} + \xi\mathbf{I} \tag{10}$$

where $\xi = 10^{-6}$ is a small constant, to ensure positive definiteness and prevent ill-conditioning, and $\mathbf{I}$ denotes the identity matrix.

During training, we observe that the GED loss tends to diminish when handling large objects, leading to near-zero gradients and hindering effective weight updates. To address this, we initially incorporate the L_1 loss to stabilize the regression of large object sizes and alleviate the learning bottleneck caused by vanishing gradients. However, the L_1 loss can produce unstable updates when the residual error is large. Inspired by Faster R-CNN [1], we replace the L_1 loss with the Smooth L_1 loss, which behaves as a quadratic function for small errors and a linear function for large errors. This design provides a more balanced gradient distribution, effectively preventing gradient vanishing for large objects while avoiding instability caused by excessive residuals during training.

The overall loss function is as follows:

$$L_{\mathrm{det}} = L_k + \lambda_{\mathrm{ged}} L_{\mathrm{ged}} + \lambda_{\mathrm{reg}} L_{\text{smooth-}L_1} \tag{11}$$

where L_k represents the classification loss, implemented as the focal loss. The hyperparameters are set as $\lambda_{\mathrm{ged}} = 2$ and $\lambda_{\mathrm{reg}} = 0.5$.

IV. Experiments and Analysis

In this section, datasets, evaluation metrics, and implementation details are first introduced. Then, comparison experiments from qualitative and quantitative perspectives demonstrate the superiority of the proposed approach over state-of-the-art methods. Subsequently, ablation studies are conducted to verify the effectiveness of the proposed modules, loss functions, and key hyperparameters. Finally, we discuss the limitations of the method to provide insights for future work.

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A. Datasets

VisDrone [6] contains 10 209 high-resolution images (about 2000×1500) with ten categories of objects, including pedestrian, people, bicycle, car, van, truck, tricycle, awning tricycle, bus, and motor. The 6471 images are used for training, 548 for validation, and 3190 for test. Since the evaluation server has been closed, following [10] and [11], the validation set is used for our evaluation.

UAVDT [40] contains 38 327 images with an average resolution of 1080×540 pixels. 23 258 images for training and 15 069 images for testing. Three categories of objects, including car, bus, and truck, are included in this dataset.

DTOD [41] contains 11 600 high-resolution remote sensing images with two categories: ships and vehicles. It includes 9280 training images and 2320 test images, totaling over 1 million annotated instances. Images vary in resolutions, up to 4700×2700, with most containing over 100 objects and the densest image including 11 112 instances.

B. Evaluation Metrics

Following the evaluation protocol on the MS COCO dataset [4], mAP, AP_{50}, and AP_{75} are used as our evaluation metrics to measure the precision. Specifically, mAP is computed by averaging over all the categories. AP_{50} and AP_{75} are calculated at the single IoU threshold 0.5 and 0.75 over all the categories. In addition, the average precision of each object category is also reported to measure the performance of each class. Moreover, for measuring the performance of different object scales, three metrics, i.e., AP_{small}, AP_{medium}, and AP_{large} are adopted. The efficiency is measured by the number of images required to be processed by the detector and the average time to process a global image and the local rescaled image crops. Specifically, the number of images refers to the summation of global images and rescaled image crops, which is denoted as #img in the subsequent experiments.

C. Implementation Details

Our proposed approach is implemented using the open-source MMDetection toolbox[1] CenterNet [17] with Hourglass-104 [42] backbone as the baseline method. Unless specified, default configurations are adopted for all the experiments. The model is trained for 160 epochs using a single NVIDIA L40 GPU with a batch size of 2, an initial learning rate of 0.01, and the input resolution of 1024×640, using the stochastic gradient descent (SGD) optimizer with momentum of 0.9 and weight decay of 0.0001.

D. Experimental Results and Analysis

Tables I–III present the comparison results of our proposed YOLD against state-of-the-art approaches on the three datasets. The results demonstrate that YOLD consistently outperforms existing methods across all three datasets, achieving stable and significant improvements. Specifically, on the VisDrone dataset [6], both one-stage detectors (e.g., CenterNet [17]) and two-stage detectors (e.g., Faster R-CNN [1]) are suboptimal due to the presence of a large number of tiny objects and severe nonuniform data distribution in aerial images. In contrast, YOLD leverages an anchor-free location map to accurately localize dense regions and perform adaptive region-of-interest cropping. This enables the detector to better focus on tiny objects while reducing background noise. Our method achieves a best AP of 38.3, surpassing the baseline by 10.5%. The gains are especially significant for small objects, while improvements for medium and large objects are marginal, which aligns with our goal of improving small object detection. Moreover, YOLD achieves superior performance in terms of the number of processed images (#img) and inference speed (s/img), indicating a favorable tradeoff between accuracy and efficiency.

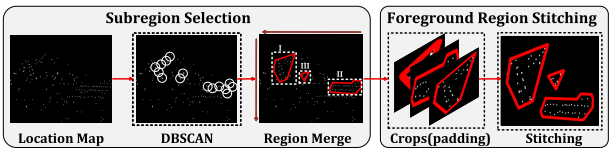


Fig. 4. Flowchart of DRE.

As shown in Table II, similar conclusions can be drawn on the UAVDT dataset [40]. General detectors struggle with severe scale variation and dense object distributions in UAV scenes, whereas YOLD achieves an AP of 20.5, significantly outperforming existing methods. Moreover, it demonstrates consistent improvements across small, medium, and large objects, further validating the effectiveness of the proposed approach.

Table III shows our detection results on the DTOD dataset [41], a large-scale benchmark for dense tiny object detection in satellite imagery. The average absolute object size in DTOD is less than 13 pixels. YOLD achieves the best performance, further validating its effectiveness, generalization capability, and superiority in detecting extremely tiny objects.

We further analyze the class-wise AP results on the VisDrone dataset [6], as detailed in Table IV. Our proposed YOLD demonstrates significantly superior detection performance compared to existing mainstream detectors, particularly excelling in small and densely packed categories such as pedestrian, person, bicycle, and awning tricycle. This performance gain can be primarily attributed to our innovative modules.

Compared with baseline detectors (e.g., RetinaNet [23], Faster R-CNN [1], and Cascade RCNN [20]) and various methods designed to address the long-tail problem (e.g., MMF [68], SimCal [69], BGS [70], and DSHNet [67]), YOLD consistently achieves superior accuracy across most categories, even without well-designed strategies for long-tail effect, which validates its generality and effectiveness in complex real-world environments. Furthermore, compared with DETR-like methods and the YOLO series, our approach still demonstrates superior performance, highlighting its effectiveness in handling densely packed and tiny objects.

Furthermore, we visualize the detection results on the three datasets in Fig. 5. These results show that our method achieves

[1]https://github.com/open-mmlab/mmdetection

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TABLE I

COMPARISON OF VARIOUS METHODS ON VISDRONE DATASET [6]."*o*," "*c*," "CA," AND "AUG," RESPECTIVELY, STAND FOR THE ORIGINAL VALIDATION SET, EVENLY IMAGE PARTITION (EIP) CROPPED IMAGES, CLUSTER-AWARE CROPPED IMAGES, AND AUGMENTED IMAGES

Method	Year & Venue	Baseline	Backbone	Test Data	#Img	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l	s/img
FRCNN [1]+FPN [43]+EIP	2015 NIPS	–	ResNet50	c	3,288	21.1	44.0	18.1	14.4	30.9	30.0	0.330
FRCNN [1]+FPN [43]+EIP	2015 NIPS	–	ResNet101	c	3,288	23.5	46.1	21.1	17.1	33.9	29.1	0.444
FRCNN [1]+FPN [43]+EIP	2015 NIPS	–	ResNeXt101	c	3,288	24.4	47.8	21.8	17.8	34.8	34.3	0.936
YOLOv9 [44]	2024 ECCV	-	GELAN	o	548	25.2	42.0	–	–	–	–	–
YOLOv10 [45]	2024 NeurIPS	-	GELAN	o	548	26.3	43.1	–	–	–	–	–
YOLOv11 [46]	2024 arXiv	-	C2K3	o	548	25.9	43.1	–	–	–	–	–
ClusDet [11]	2019 ICCV	FRCNN+FPN	ResNet50	o+ca	2,716	26.7	50.6	24.7	17.6	38.9	51.4	0.273
ClusDet [11]	2019 ICCV	FRCNN+FPN	ResNet101	o+ca	2,716	26.7	50.6	25.2	17.2	39.3	54.9	0.266
ClusDet [11]	2019 ICCV	FRCNN+FPN	ResNeXt101	o+ca	2,716	28.4	53.2	26.4	19.1	40.8	51.4	0.773
DMNet [12]	2020 CVPRW	FRCNN+FPN	ResNet50	o+ca	2,736	28.2	47.6	28.9	19.9	39.6	55.8	0.29
DMNet [12]	2020 CVPRW	FRCNN+FPN	ResNet101	o+ca	2,736	28.5	48.1	29.4	20.0	39.7	57.1	0.36
DMNet [12]	2020 CVPRW	FRCNN+FPN	ResNeXt101	o+ca	2,736	29.4	49.3	30.6	21.6	41.0	56.7	0.61
AMRNet [47]	2020 Arxiv	FRCNN+FPN	ResNet50	o+aug	–	31.7	–	–	23.0	43.4	58.1	–
AMRNet [47]	2020 Arxiv	FRCNN+FPN	ResNet101	o+aug	–	31.7	–	–	29.0	45.5	**60.9**	–
AMRNet [47]	2020 Arxiv	FRCNN+FPN	ResNeXt101	o+aug	–	32.1	–	–	23.2	43.9	60.5	–
GLSAN [15]	2021 TIP	FRCNN+FPN	ResNet50	o+ca	30.7	55.4	30.0	–	–	–	–	–
GLSAN [15]	2021 TIP	FRCNN+FPN	ResNet101	o+ca	30.7	55.4	30.0	–	–	–	–	–
GLSAN [15]	2021 TIP	Cascade R-CNN	ResNet50	o+ca	32.5	55.8	30.0	–	–	–	–	–
PRDet [48]	2023 TCSVT	FRCNN+FPN	ResNet50	o+ca	1644	31.1	52.7	32.1	24.7	39.8	51.2	0.126
PRDet [48]	2023 TCSVT	FRCNN+FPN	ResNet101	o+ca	1644	31.7	53.3	33.2	25.1	40.8	50.9	0.144
PRDet [48]	2023 TCSVT	FRCNN+FPN	ResNeXt101	o+ca	1644	32.0	53.9	33.2	25.6	40.8	52.9	0.195
AFSM [49]	2020 Arxiv	CenterNet	CBResNet50	–	–	33.95	60.46	32.69	–	–	–	–
AFSM [49]	2020 Arxiv	CenterNet	CBResNet50+DCN	–	–	35.43	61.88	34.60	–	–	–	–
FCOS+SAHI [50]	2022 ICIP	FCOS	–	ca	–	–	38.5	–	–	–	–	–
VFNet+SAHI [50]	2022 ICIP	VFNet	–	ca	–	–	42.2	–	–	–	–	–
TOOD+SAHI [50]	2022 ICIP	TOOD	–	ca	–	–	43.5	–	–	–	–	–
QueryDet [51]	2022 CVPR	RetinaNet	ResNet50	o	548	28.32	48.14	28.75	–	–	–	0.364
CEASC [52]	2023 CVPR	RetinaNet	ResNet50	o	548	28.8	35.0	27.7	–	–	–	0.070
CEASC [52]	2023 CVPR	GFL	ResNet18	o	548	28.7	50.7	28.4	–	–	–	**0.046**
DETR [53]	2020 ECCV	DETR	ResNet50	o	548	24.1	40.1	–	–	–	–	–
Deformable DETR [54]	2020 ICLR	DETR	ResNet50	o	548	27.1	42.2	–	–	–	–	–
RT-DETR [55]	CVPR 2024	DETR	ResNet50	o	548	28.4	47.0	–	–	–	–	–
DQ-DETR [56]	2024 ECCV	DETR	ResNet50	o	548	37.0	60.9	37.9	–	–	–	–
YOLC [16]	2024 T-ITS	CenterNet	ResNet50	o+ca	1644	31.8	55.0	31.7	24.7	42.3	45.0	0.441
YOLC [16]	2024 T-ITS	CenterNet	ResNeXt101	o+ca	1644	33.7	57.4	33.8	26.1	44.9	52.8	0.543
YOLC [16]	2024 T-ITS	CenterNet	HRFPN	o+ca	2192	37.8	61.6	39.4	30.6	48.1	54.1	0.716
CenterNet (Baseline) [17]	2019 Arxiv	–	Hourglass104	o	548	27.8	47.9	27.6	21.3	42.1	49.8	0.215
CenterNet [17]	2019 Arxiv	–	ResNet50	o	548	23.9	45.5	22.1	14.4	36.3	44.9	0.073
CenterNet [17]	2019 Arxiv	CenterNet	HRFPN	o	548	29.1	49.2	30.1	22.3	43.9	52.4	0.179
YOLD (Ours)	–	CenterNet	ResNet50	o+ca	1096	32.1	56.2	32.3	25.0	43.9	47.8	0.432
YOLD (Ours)	–	CenterNet	ResNet101	o+ca	1096	35.6	57.8	35.9	27.7	44.6	49.4	0.447
YOLD (Ours)	–	CenterNet	ResNeXt101	o+ca	1096	36.4	59.2	36.8	28.3	46.9	50.1	0.451
YOLD (Ours)	–	CenterNet	HRFPN	o+ca	1096	**38.3**	**62.4**	**39.7**	**31.1**	**48.3**	53.7	0.545

promising detection accuracy across various aerial scenarios, particularly excelling at detecting numerous tiny objects that are easily missed by general detectors, as well as obtaining reliable results for nonuniformly distributed objects. Furthermore, as illustrated in Fig. 6, compared with the baseline method, i.e., CenterNet [17], our proposed YOLD produces more accurate detections in dense regions and demonstrates remarkable advantages in tiny object detection.

E. Ablation Study

To validate the effectiveness of the proposed components, we conduct a series of ablation experiments on the VisDrone dataset [6], covering LMG module, DRE module, GED loss function designs, and hyperparameter configurations.

1) Effect of GDTM: To precisely regress the object centers, we replace the conventional heatmap in the baseline model with our proposed Gaussian-based distance transform map. As shown in Table V, this modification leads to a notable performance gain of 7% AP. To further analyze the effectiveness of the adaptive parameter γ in GDTM, we visualize the response–distance curves in Fig. 7. The results indicate that regions with different densities correspond to distinct γ responses: the denser the neighborhood, the sharper the GDTM decay. Moreover, as γ increases with local density, the adaptive strategy produces sharper peaks to resolve dense objects while maintaining broader responses in sparse regions to ensure sufficient supervision. This dynamic adjustment enhances localization precision in both dense and sparse

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TABLE II
PERFORMANCE COMPARISON ON UAVDT [40]

Method	Baseline	Backbone	#Img	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
R-FCN [21]	–	ResNet50	15,069	7.0	17.5	3.9	4.4	14.7	12.1
SSD [3]	–	–	15,069	9.3	21.4	6.7	7.1	17.1	12.0
FRCNN [1]	–	–	15,069	5.8	17.4	2.5	3.8	12.3	9.4
FRCNN [1]+FPN [43]	–	–	15,069	11.0	23.4	8.4	8.1	20.2	26.5
FRCNN [1]+FPN [43]+EIP	–	–	60,276	6.6	16.8	3.4	5.2	13.0	17.2
ClusDet [11]	FRCNN+FPN	ResNet50	25,427	13.7	26.5	12.5	9.1	25.1	31.2
DMNet [12]	FRCNN+FPN	ResNet50	32,764	14.7	24.6	16.3	9.3	26.2	35.2
DREN [14]	MaskRCNN+FPN	ResNet50	–	15.1	–	–	–	–	–
DREN [14]	MaskRCNN+FPN	ResNet101	–	17.1	–	–	–	–	–
AMRNet [47]	FRCNN+FPN	ResNet50	–	18.2	30.4	19.8	10.3	31.3	33.5
GLSAN [15]	FRCNN+FPN	ResNet50	–	17.0	28.1	18.8	–	–	–
GLSAN [15]	FRCNN+FPN	ResNet101	–	17.1	28.3	18.8	–	–	–
CEASC [52]	GFL	–	15,069	17.1	30.9	17.8	–	–	–
CenterNet [17] (Baseline)	–	Hourglass104	15,069	13.2	26.7	11.8	7.8	26.6	13.9
YOLC [16]	CenterNet	HRFPN	30,138	19.3	30.9	20.1	10.9	32.2	35.5
YOLD (Ours)	CenterNet	HRFPN	30,138	**20.5**	**31.4**	**20.7**	**11.4**	**32.8**	**36.1**



Fig. 5. Visualization of our YOLD detection results on VisDrone [47] (first row), UAVDT [48] (second row), and DTOD (third row).

scenarios. We also provide a comparative visualization with the conventional heatmap to further illustrate the improvement, as shown in Fig. 8.

2) Effect of DRE: To address the nonuniform object distribution problem, we introduce the DRE module and evaluate its impact using different clustering strategies. As shown in Table VI, using traditional K-means yields a baseline performance of 33.2 AP. When replacing K-means with DBSCAN, the performance improves significantly by 2.4 points, demonstrating the advantage of density-aware clustering in discovering effective object clusters. Furthermore, we incorporate our proposed FRS strategy, which concatenates cropped foreground regions to enhance context for dense areas. With FRS, the performance further increases by 0.4 points, reaching 36.0 AP. These results validate the effectiveness of the DRE module and highlight the importance of adaptive clustering and region aggregation in improving detection accuracy under complex spatial distributions.

3) Effect of GED: To further improve the detection accuracy for small and dense objects, we investigate several distance-based loss functions, as summarized in Table VII. Starting from a baseline model using the standard L_1 loss, we replace it with GIoU [71] and GWD [72] losses. Both alternatives show steady improvements, with GWD achieving 33.8 AP, outperforming GIoU by 0.4 points.

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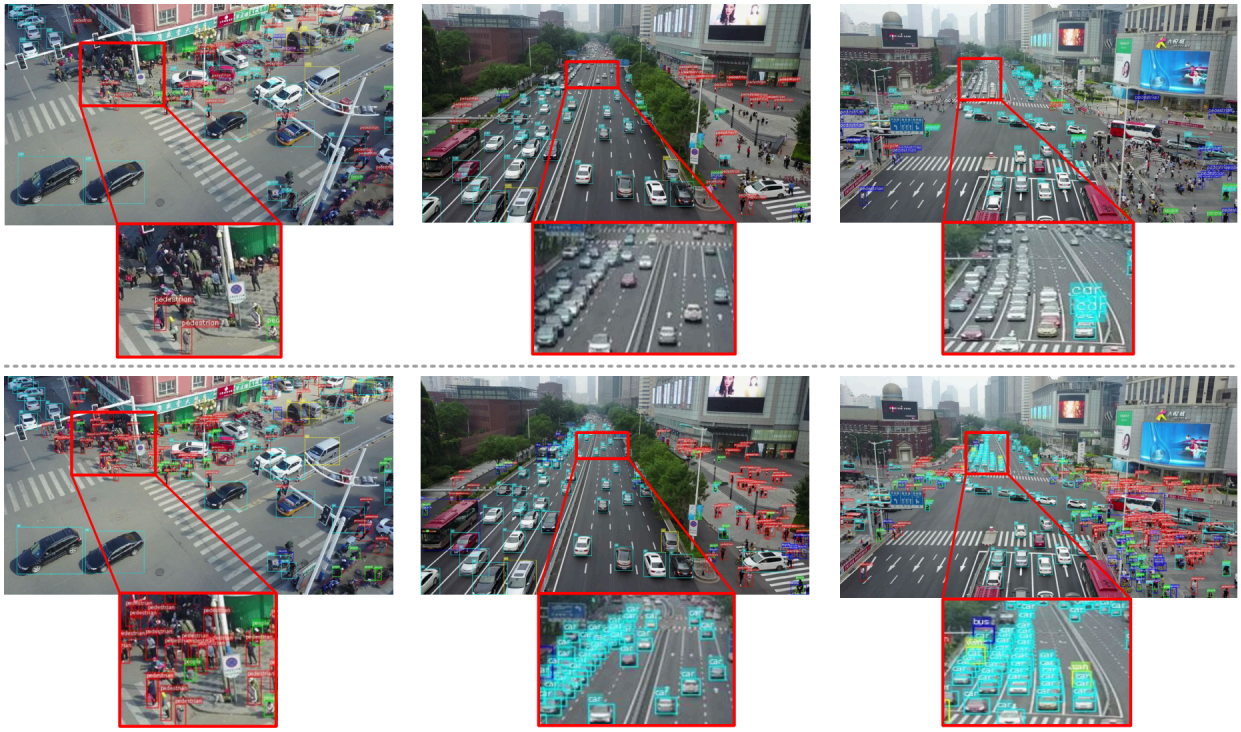


Fig. 6. Visualization of CenterNet (top) and our YOLD (bottom) detection results on VisDrone.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DTOD [41] TEST SET

Method	Backbone	AP_{50}	AP_{75}
FRCNN [1]	ResNet50	11.6	2.1
Cascade R-CNN [20]	ResNet50	9.9	1.7
YOLOv3 [22]	Darknet53	4.4	0.1
FCOS [25]	ResNet50	7.0	1.2
RepPoints [57]	ResNet50	5.0	0.8
FoveaBox [24]	ResNet50	12.0	1.8
ATSS [58]	ResNet50	14.6	2.2
Dynamic R-CNN [58]	ResNet50	7.1	2.0
Dyhead [59]	ResNet50	10.5	1.6
TPH-YOLOv5 [60]	CSPDarknet-s	10.1	1.9
DDOD [61]	ResNet50	14.5	2.2
YOLOv7 [62]	CSPDarkNet-X	9.2	1.5
DINO [63]	ResNet50	8.5	1.1
YOLO-MS [64]	CSPDarkNet-s	5.9	1.7
DynamicDet [65]	CSPDarkNet-X	11.5	1.6
YOLOv5-19 [66]	CSPDarkNet-s	13.8	2.4
SCDNet [41]	CSPDarkNet-s	24.2	3.3
YOLD (Ours)	HRFPN	**25.1**	**3.6**

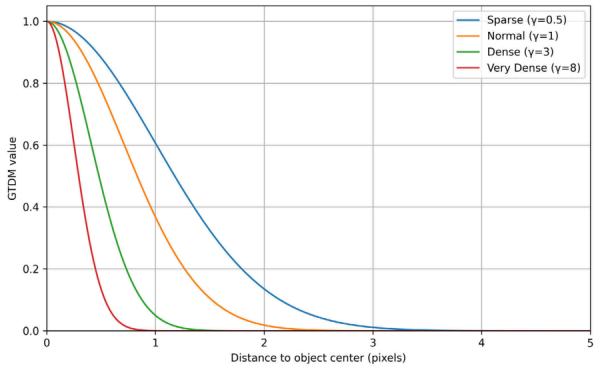


Fig. 7. Response curves of the proposed GDTM under different γ values. Higher γ values (dense regions) cause a sharper decline in response to enhance localization precision, while lower values (sparse regions) maintain a broader response to provide sufficient supervision.

To better capture the shape and spatial relationship of dense objects, we introduce MD and GED as regression losses. Since MD is a special case of GED with covariance normalization, we compare them separately. As shown in Table VII, using MD alone yields an AP of 34.1, while combining it with L_1 loss (MD $+L_1$) further improves the AP to 34.5. Similarly, GED alone achieves 34.2 AP, and its combination with L_1 (GED $+L_1$) reaches the best performance of 34.7 AP, with significant gains in both AP_{50} and AP_{75}.

However, we observe that the GED-based loss may produce insufficient gradients when regressing large objects, which can lead to unstable bounding-box refinement. Inspired by the loss function design in Faster R-CNN, we replace the standard L_1 loss with Smooth L_1 to achieve a more balanced and stable optimization. As shown in Table VIII, Smooth L_1 consistently outperforms L_1 across all object scales. Notably, the improvements on AP_m and AP_l indicate enhanced regression stability for medium and large objects, which are more sensitive to gradient attenuation in GED-based optimization.

4) Effect of FRS: We investigate the effect of different FRS strategies on detection performance, as summarized in Table IX. Directly feeding cropped patches into the detector without stitching achieves an accuracy of 35.6 AP with an inference speed of 0.723 s/img. Stitching both the cropped patches and the original image

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TABLE IV

DETECTION PERFORMANCE OF EACH CLASS ON VISDRONE [6] VALIDATION SET.PED. AND AWN. ARE SHORT FOR PEDESTRIAN AND AWNING TRICYCLE. RS IS SHORT FOR RANDOM SAMPLER

Method	Backbone	Ped.	People	Bicycle	Car	Van	Truck	Tricycle	Awn.	Bus	Motor
Comparison with base models											
RetinaNet+RS [23]	ResNet50	13.0	7.9	1.4	45.5	19.9	11.5	6.3	4.2	17.8	11.8
FRCNN+RS [1]	ResNet50	21.4	15.6	6.7	51.7	29.5	19.0	13.1	7.7	31.4	20.7
FRCNN+RS [1]	ResNet101	20.9	14.8	7.3	51.0	29.7	19.5	14.0	8.8	30.5	21.2
FRCNN+RS [1]	ResNeXt101	21.3	15.5	7.9	53.0	29.5	20.5	14.7	8.9	32.1	21.6
CRCNN+RS [20]	ResNet50	22.2	14.8	7.6	54.6	31.5	21.6	14.8	8.6	34.9	21.4
RetinaNet+DSHNet [67]	ResNet50	14.1	8.9	1.3	48.2	24.8	14.2	8.8	6.0	21.6	13.1
FRCNN+DSHNet [67]	ResNet50	22.5	16.5	10.1	52.8	32.6	22.1	17.5	8.8	39.5	23.7
FRCNN+DSHNet [67]	ResNet101	21.7	16.0	10.1	52.2	31.6	22.7	17.1	9.5	38.6	24.0
FRCNN+DSHNet [67]	ResNeXt101	23.3	16.7	11.4	53.7	33.1	23.8	19.5	11.1	40.0	25.5
CRCNN+DSHNet [67]	ResNet50	23.2	16.1	11.2	55.5	33.5	25.2	19.1	10.0	43.0	25.1
Comparison with solutions to long-tail problems											
FRCNN+RS+MMF [68]	ResNet50	21.6	15.3	9.6	51.5	28.5	20.4	15.9	7.5	33.7	21.6
FRCNN+SimCal [69]	ResNet50	18.7	13.8	5.7	48.2	26.4	16.4	13.6	5.9	27.0	19.4
FRCNN+RS+BGS [70]	ResNet50	21.8	16.0	9.1	51.8	31.1	19.8	15.0	8.4	36.1	21.5
FRCNN+DSHNet [67]	ResNet50	22.5	16.5	10.1	52.8	32.6	22.1	17.5	8.8	39.5	23.7
Based on the SOTA cropping method											
DMNet (FRCNN+RS) [12]	ResNet50	28.1	19.7	13.3	57.3	36.1	24.8	20.1	12.0	42.9	26.4
DMNet [12]cropping+DSHNet [67]	ResNet50	28.5	20.4	15.9	56.8	37.9	30.1	22.6	14.0	47.1	29.2
YOLC [16]	HRFPN	39.5	26.5	23.6	65.3	45.2	**36.5**	28.6	**19.7**	**58.8**	35.5
CenterNet [17] (Baseline)	Hourglass104	28.7	18.5	12.7	57.9	37.6	22.9	18.4	11.1	43.2	20.9
YOLD	HRFPN	**39.6**	**27.2**	**24.1**	**65.7**	**46.3**	35.3	**29.8**	**19.7**	58.4	**35.9**

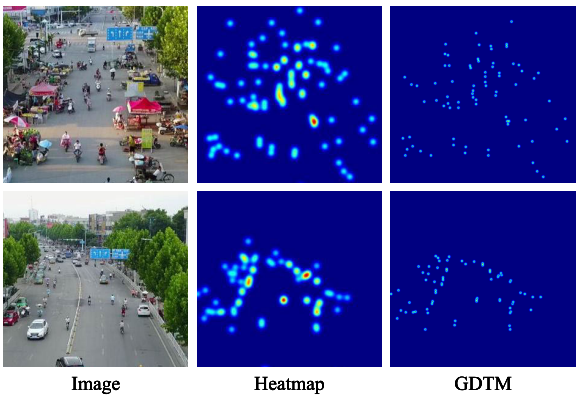


Fig. 8. Rows 1 and 2, from left to right, there are original images, predicted heatmaps, and predicted GDTM (trained with the same network). It can be seen that objects of the predicted GDTM are distinguishable, and the predicted heatmaps show severe overlaps.

TABLE V

EFFECT OF THE GDTM ON DETECTION PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Heatmap (CenterNet [17])	29.1	49.2	30.1	22.3	43.9	52.4
GDTM (Ours)	**36.1**	**60.4**	**36.7**	**27.6**	**48.1**	**57.1**

TABLE VI

EFFECT OF THE DRE MODULE ON DETECTION PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
K-means	33.2	54.2	31.1	22.5	39.9	48.4
DBSCAN	35.6	60.0	36.2	27.8	47.0	49.7
DRE (DBSCAN+FRS)	**36.0**	**60.3**	**36.5**	**27.9**	**47.5**	**50.5**

improves the accuracy to 36.4 AP; however, the increased number of images substantially reduces the inference speed to 0.964 s/image. Finally, feeding only the stitched image without individual cropped patches achieves a slightly lower accuracy of 36.0 AP but a faster inference speed of 0.427 s/img. These results indicate a clear tradeoff between detection accuracy and inference efficiency. While stitching both cropped patches and the original image can achieve the highest accuracy, the associated computational cost and reduced inference speed make it impractical for real-time or large-scale applications. Therefore, in practice, a balanced strategy, feeding only the stitched image without individual cropped patches, is preferred to maintain an acceptable compromise between accuracy and efficiency.

TABLE VII

PERFORMANCE COMPARISON UNDER DIFFERENT REGRESSION LOSSES

Method	AP	AP_{50}	AP_{75}
Baseline (L_1)	33.0	58.2	33.1
GIoU [71]	33.4	58.0	33.8
GWD [72]	33.8	58.3	34.3
MD	34.1	58.5	34.5
MD + L_1	34.5	59.1	34.7
GED	34.2	58.4	33.9
GED + L_1	34.7	59.2	35.0
GED + Smooth L_1	**35.0**	**59.8**	**36.2**

TABLE VIII

EFFECT OF REGRESSION LOSS DESIGN ON DETECTION PERFORMANCE

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
L_1	34.7	59.2	35.0	28.6	43.9	50.4
smooth L_1	**35.0**	**59.8**	**36.2**	**28.8**	**47.0**	**50.9**

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TABLE IX

EFFECT OF THE FRS MODULE ON DETECTION PERFORMANCE. "*C*" DENOTES THE USE OF CROPPED PATCHES, WHILE "*S*" DENOTES THE USE OF STITCHED IMAGE

Method	AP	AP_{50}	AP_{75}	#img	s/img
FRS (w/-C)	35.6	60.0	36.2	2192	0.723
FRS (w/-C, -S)	**36.4**	**60.8**	**36.6**	2740	0.964
FRS (w/-S)	36.0	60.3	36.5	1096	0.427

TABLE X

EFFECT OF DETECTION HEAD ENHANCEMENT ON DETECTION PERFORMANCE

Detection Head	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Baseline detection head	35.0	59.3	36.2	28.5	45.6	50.9
Enhanced detection head	**35.7**	**60.0**	**36.8**	**29.1**	**45.8**	**51.3**

5) Effect of Enhanced Detection Head: To address the limitation of the baseline detection head in handling small objects, we enhance the head architecture by incorporating ASPP [34] and CBAM [35] modules. These modifications enable the model to better capture fine-grained details and contextual cues, which are crucial for accurately detecting small objects. As shown in Table X, the enhanced detection head consistently improves performance across all evaluation metrics, achieving an AP gain from 35.0 to 35.7. Notably, AP_s is further improved by 0.6%, indicating that this technique is particularly effective for detecting small objects.

6) Different Backbones: To evaluate the generalization ability of our approach and ensure a fair comparison with prior clustering-based detectors, we examine several commonly used backbone networks, including ResNet50, ResNet101, ResNeXt101, and HRFPN. As shown in Table I, replacing the original Hourglass backbone in CenterNet with HRFPN yields a 1.3% AP improvement (27.8 → 29.1), confirming the importance of maintaining high-resolution feature representations for tiny object detection in aerial scenes. When integrating into these backbones, our YOLD framework consistently delivers substantial gains. Specifically, YOLD improves the AP of the ResNet50 backbone by + 8.2% compared with the baseline (23.9 → 32.1). Moreover, adopting stronger backbones such as ResNet101 and ResNeXt101 further improves the overall detection performance, demonstrating that our method can effectively benefit from enhanced feature representations across different backbone families. Notably, YOLD with HRFPN obtains the best overall performance, reaching 38.3 AP, which also demonstrates the same conclusion in YOLC. Therefore, we adopt HRFPN as the default backbone in our subsequent experiments.

7) Neighborhood Radius ϵ in DBSCAN: To evaluate the impact of the neighborhood radius parameter ϵ on detection performance, we conduct ablation experiments with four settings: fixed values (30, 40, and 50) and an adaptive setting (dynamic ϵ). The fixed values are selected based on the empirical distribution of pairwise Euclidean distances between localization points, as illustrated in Fig. 9, where a clear peak appears around 40 pixels, indicating a representative spatial scale of dense object neighborhoods. Accordingly, $\epsilon = 40$ is chosen as the central fixed setting, while $\epsilon = 30$ and $\epsilon = 50$ are included to examine underclustering and overclustering scenarios, respectively. As shown in Table XI, the adaptive dynamic ϵ consistently outperforms all fixed values across all evaluation metrics, achieving an AP of 38.3. This demonstrates that the adaptive ϵ effectively accommodates varying object densities across different scenes, leading to improved detection accuracy.

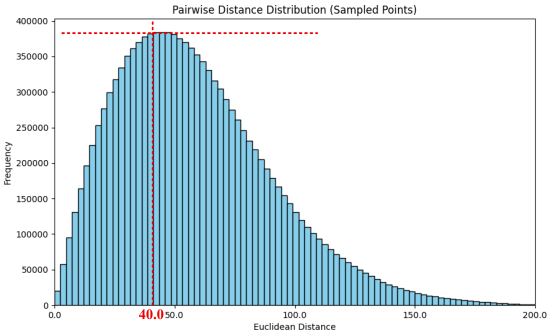


Fig. 9. Neighborhood radius ϵ in DBSCAN.

TABLE XI

EFFECT OF DIFFERENT NEIGHBORHOOD RADIUS ϵ SETTINGS IN DBSCAN ON DETECTION PERFORMANCE

ϵ	AP	AP_{50}	AP_{75}
30	36.4	60.8	37.7
40	37.9	62.1	38.9
50	37.6	61.2	38.3
Dynamic ϵ	**38.3**	**62.4**	**39.7**

TABLE XII

ABLATION STUDY ON THE SCALING FACTOR α IN THE GLOBAL DECAY MECHANISM. A LARGER α INCREASES THE GLOBAL DECAY STRENGTH, AFFECTING THE RESPONSE INTENSITY FOR DENSE OBJECT REGIONS

Method	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
$\alpha = 2$	29.1	49.2	30.1	22.3	43.9	52.4
$\alpha = 5$	35.5	59.3	35.9	26.9	47.8	59.4
$\alpha = 10$	**36.1**	**60.4**	**36.7**	**27.6**	**48.1**	**57.1**
$\alpha = 15$	35.2	58.8	35.2	26.1	47.6	55.8

8) Scaling Factor \alpha: We conduct an ablation study on the scaling factor α used in the GDTM to investigate its impact on the regression quality for dense and small objects. As shown in Table XII, setting $\alpha = 2$ leads to a relatively slow decay, resulting in suboptimal performance with 29.1 AP. Increasing α to 5 significantly boosts the AP to 35.5, indicating that faster decay helps suppress noise and enhances the response in dense regions. The best performance is achieved at $\alpha = 10$, reaching 36.1 AP, 60.4 AP_{50}, and 36.7 AP_{75}, as well as notable gains across all object scales. However, further increasing α to 15 leads to a slight performance drop, suggesting that overly fast decay may suppress useful information. These results demonstrate that a moderate scaling factor strikes a good balance between noise suppression and

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(a) (b)

Fig. 10. Comparison of detection results of the proposed YOLD on the VisDrone [6] validation set. (a) Typical failure cases of the original YOLD, where green boxes denote classification errors and pink boxes indicate missed detections caused by severe occlusion. (b) Detection results of YOLD with the CBAM [35] module integrated.

TABLE XIII
EFFECT OF DIFFERENT MODULES ON DETECTION PERFORMANCE

Method	people	pedestrian
YOLD	39.6	27.2
YOLD + DCN [73]	39.8(+0.2)	27.3(+0.1)
YOLD + CBAM [35]	40.0(+0.4)	27.6(+0.4)
YOLD + ASPP [34]	39.6(+0.0)	27.4(+0.2)

response preservation and confirm the necessity of controlling the global decay strength for robust localization in dense scenes.

9) Limitations and Discussion: As shown in Fig. 10(a), we present two representative failure cases that highlight the limitations of the proposed method in complex scenarios. The pink box indicates severe occlusion, while the green box denotes category confusion, such as between "people" and "pedestrian." In the VisDrone dataset [6], category "pedestrian" refers to individuals in standing or walking postures, whereas "people" represents those in other postures. To address these challenges, we integrate several context modeling and occlusion-aware modules, including DCN [73], ASPP [34], and CBAM [35]. The corresponding ablation experiments are conducted to evaluate their effectiveness. As shown in Table XIII and qualitatively illustrated in Fig. 10(b), it demonstrates that these enhancements alleviate the challenges to a certain extent. It motivates us in the future to incorporate more advanced context modeling and adaptive feature aggregation strategies to better capture the spatial and semantic relationships between objects and their backgrounds, aiming at improving the model's robustness against complex backgrounds, severe occlusion, and interclass confusion.

Furthermore, the current YOLD framework is not fully end-to-end, as the cropping operation is performed at the image level prior to network forwarding. While image-level cropping is a common practice in current high-resolution aerial object detection, it limits the possibility of joint optimization across the entire pipeline. We plan to implement DRE and selection directly at the feature level through differentiable region selection [54], [74] or region re-grouping [75], [76], by learning density-aware feature gates or lightweight region assignment weights that allow gradients to flow through the region selection process, enabling fully end-to-end optimization.

V. CONCLUSION

In this article, we have presented an anchor-free network framework, YOLD, for object detection in aerial images. In particular, we mainly attack two intractable challenges in aerial images: a large number of tiny objects and severely nonuniform data distribution. For the former problem, we present GDTM to substitute the heatmaps in the CenterNet, a strong anchor-free object detection network. By simply substituting operation, the detection performance obtains substantial improvements, especially for small objects. For the latter problem, we further put forward a DRE module, which adaptively searches clustered object regions and rescales them to accommodate the detectors, while being lightweight and computationally efficient. Finally, we employ a GED-based regression loss as a substitute for the original regression loss in CenterNet, providing enhanced stability and differentiability while facilitating efficient model training and implementation. Extensive experiments conducted on three aerial image datasets demonstrate the effectiveness and superiority compared with state-of-the-art methods.

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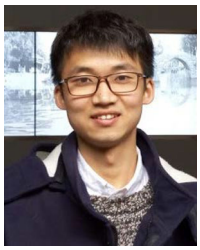
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